Helping Lonley People: Impact of Targeted Within-Community Rewiring vs Random Edge Addition on Local Clustering of Social Network Nodes

Ali Najafzadeh

May 2025

Abstract

Social isolation generally occurs among people who stand at the border of network structures [3]. This study implements two different network intervention methods to increase the quantity of the isolated individuals' relations to enhance their social life. Network interventions are (1) targeted within-community rewiring which connects each individual node to a structurally similar peer within the same community, and (2) uniform random edge addition which adds connections between randomly selected node pairs. This study uses simulation tests on Facebook social dataset to evaluate each intervention by measuring changes in the clustering coefficient (measuring how tightly people are connected in a group). Results show the targeted approach defeats random strategy in local clustering coefficient with high statistical probability $p = 2.39 \times 10^{-28}$ and a large effect size (Cohen's $d = 8.1918$). Targeted rewiring shows effectiveness even during early rounds because it enhances isolated network nodes (left-behind nodes) through two iterations while random edge additions struggles to connect them. The performance of targeted strategy slowly decrease through time because the probability to create new triangular clusters will be less. On one hand, targeted approach improves local clustering it offers limited improvements in global measurements like average shortest-path length (ASPL). On the other hand, random approach reduces ASPL but it is less beneficial to the isolated nodes. The targeted method tries to stay within community to add meaningful connections to a node rather than connecting them to a hub.

Keywords: Network science, Social networks, Local clustering coefficient, Targeted rewiring, Community structure, Loneliness

1 Introduction

1.1 Social Isolation and Loneliness in Network Structure

Loneliness is typically defined as the subjective perception of a difference between desired and actual social relationships [8]. Social isolation refers to the lack of social contacts or meaningful participation in social life but these assumptions can also have some exceptions, individuals can feel lonely despite having many social connections, or feel satisfied

despite limited contact [8]. In network science, lonely individuals have sparse connections and low local clustering and low degree [3]. Prior work indicates that increasing social participation and linking individuals to communities can reduce their loneliness [1]. The PALS study (Project About Loneliness and Social networks) also emphasised that only increasing the degree (the number of social ties) is insufficient and the quality of relationships play a crucial role in reducing loneliness [1]. Participants being introduced to meaningful supportive community networks showed improvements in well-being [1]. Assessments of loneliness reveal that individuals with large personal networks may still experience high loneliness when they are alone because structural and contextual integration is more important than just size of network [11, 5]. Rico-Uribe et al. also found that loneliness is more predictive of health outcomes than objective network characteristics like quantity of contact [11]. Individuals who feel lonely often have left behind positions in their social networks and this isolation can create a self-reinforcing cycle where the lonely become more disconnected over time [3]. To break this cycle, researchers have suggested aggressively targeting the people who are in the border of network to help repair their social networks [3].

1.2 Targeted Intervention

In this study, a targeted network intervention strategy was used to help left behind individuals integrate into their communities. Targeted within-community strategy was compared to a control strategy which is the addition of an equal number of random ties throughout the network. In the targeted strategy, one new friendship tie was added from each isolated (low-degree, low-clustering) node to the most similar person within their community. The comparison measurement was local clustering coefficient of nodes, average shortest-path length (ASPL) and the number of left-behind nodes across rounds of edge additions.

Average shortest path length (ASPL) $\langle d \rangle$ is defined as:

$$\langle d \rangle = \frac{1}{N(N-1)} \sum_{\substack{i,j \in V \\ i \neq j}} d(i,j) \tag{1}$$

where:

- N is the number of nodes in the network,
- V is the set of all nodes,
- $d(i,j)$ is the length of the shortest path between nodes i and j.

ASPL was tracked as a secondary metric to ensure that the global topology of the network was not unintentionally disrupted. ASPL is reduced when nodes are connected to hubs and such additions can undermine community boundaries and decrease local cohesion [4]. To prevent this, connections to high-degree nodes (hubs) was explicitly avoided by penalising it in the targeted strategy as these might rapidly reduce ASPL without providing meaningful integration for individuals. By using Jaccard similarity measure, and excluding hubs from recommendations, the reinforcement of local social structure was prioritised, and the small-world properties of the original network were preserved [12].

2 Methods

2.1 Dataset Description

The facebook dataset from the Stanford Network Analysis Project (SNAP) was used in this study [10]. This dataset is an anonymized ego-network data collected from Facebook with 4,039 nodes and 88,234 undirected edges, with each node representing a user and each edge showing a friendship between two users and it is collected in 2009 [10]. Small-world characteristics are shown by the dataset including a global clustering coefficient of approximately 0.605 and an average shortest path length of 3.69 [10, 12]. The graph is fully connected (a single giant component) and without self-loops or multiple edges from the same pair of nodes. This dataset was chosen because its ordinary scale allows rapid intervention and its community structure provides a meaningful testbed for measuring small-world characterstics [12].

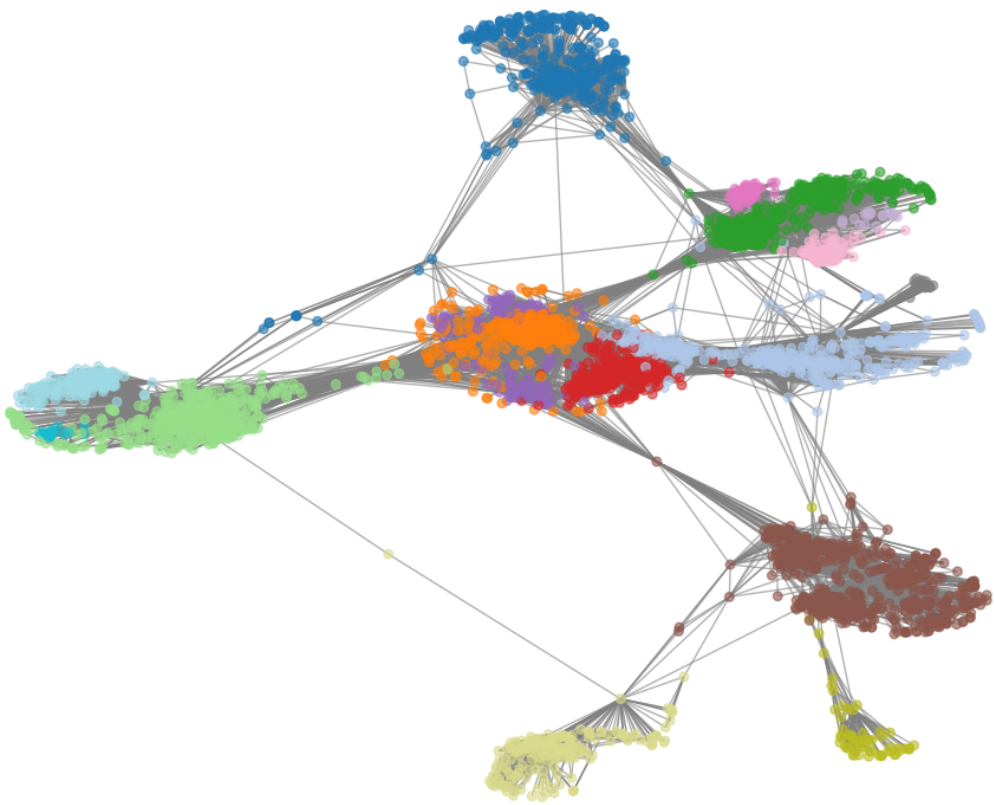

Figure 1: Community visualization of the Facebook network: Undirected graph with spring layout of the facebook_combined graph colored by community which is detected by Louvain algorithm [7]. Densely clustered groups indicate strong community structure.

2.2 Left behind Node Identification

The left behind nodes were defined as those with both low degree and low local clustering coefficient which means a few friends and low connectivity between friends (not strongly connected friends). The local clustering coefficient c_u of a node u is defined as the fraction of possible pairs of u's neighbors that are actually connected to each other. In other words, $c_u = 1.0$ (100%) means that all of u's friends know each other (making a clique), whereas $c_u = 0$ shows none of u's friends are friends with each other. Nodes with low degree automatically have $c_u = 0$ because there would be no triangles and the focus is on them in this study.

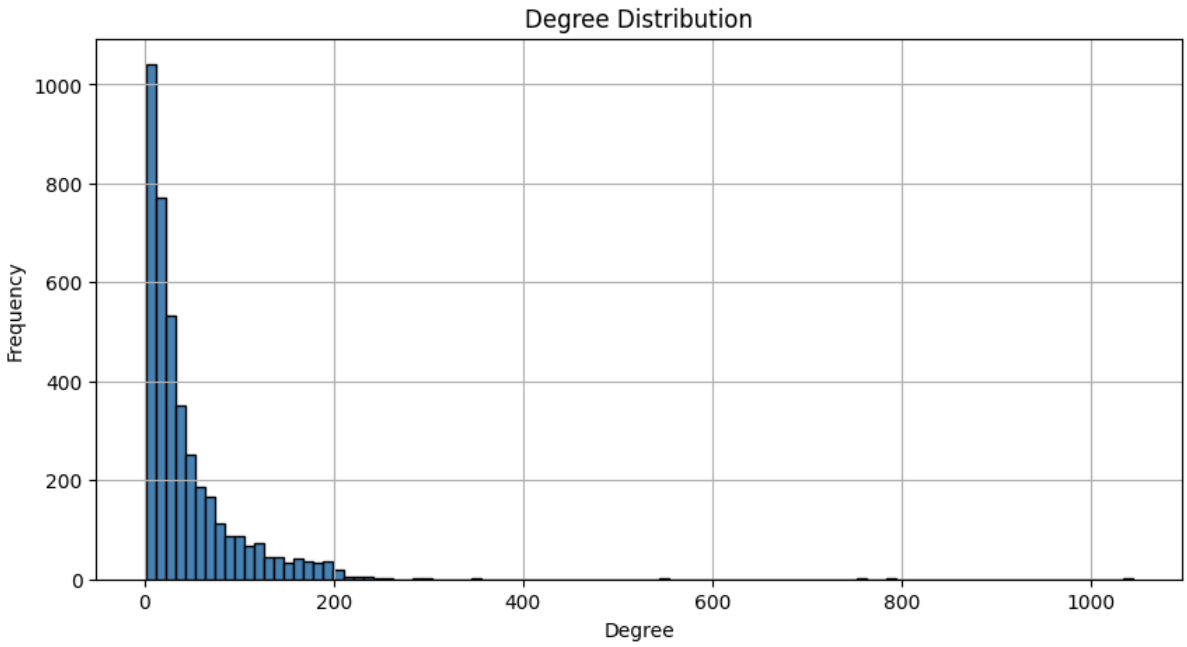

Figure 2: Degree distribution of full network: Histogram of node degrees in the facebook_combined graph. The heavy-tailed distribution reflects the presence of hubs and many low-degree nodes, consistent with real-world social networks [6].

The **local clustering coefficient** C_v of a node v is defined as:

$$C_v = \frac{2e_v}{k_v(k_v - 1)}$$

where:

- k_v is the degree of node v,
- e_v is the number of edges between the neighbors of v.

The **global clustering coefficient** C (also called transitivity) is defined as:

$$C = \frac{3 \times \text{number of triangles}}{\text{number of connected triples of nodes}} \tag{2}$$

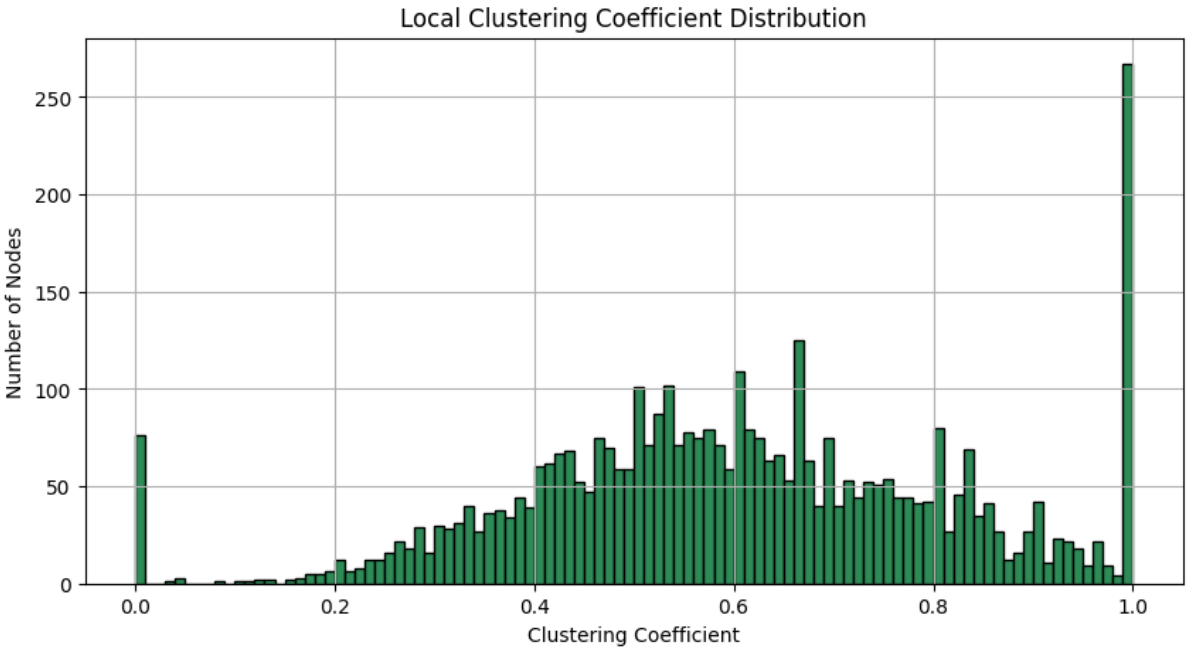

Figure 3: Local clustering coefficient distribution: Distribution of local clustering coefficients for all nodes. The bimodal shape with a peak at 1.0 reflects the presence of many fully embedded individuals (cliques), while lower values indicate sparsely connected nodes.

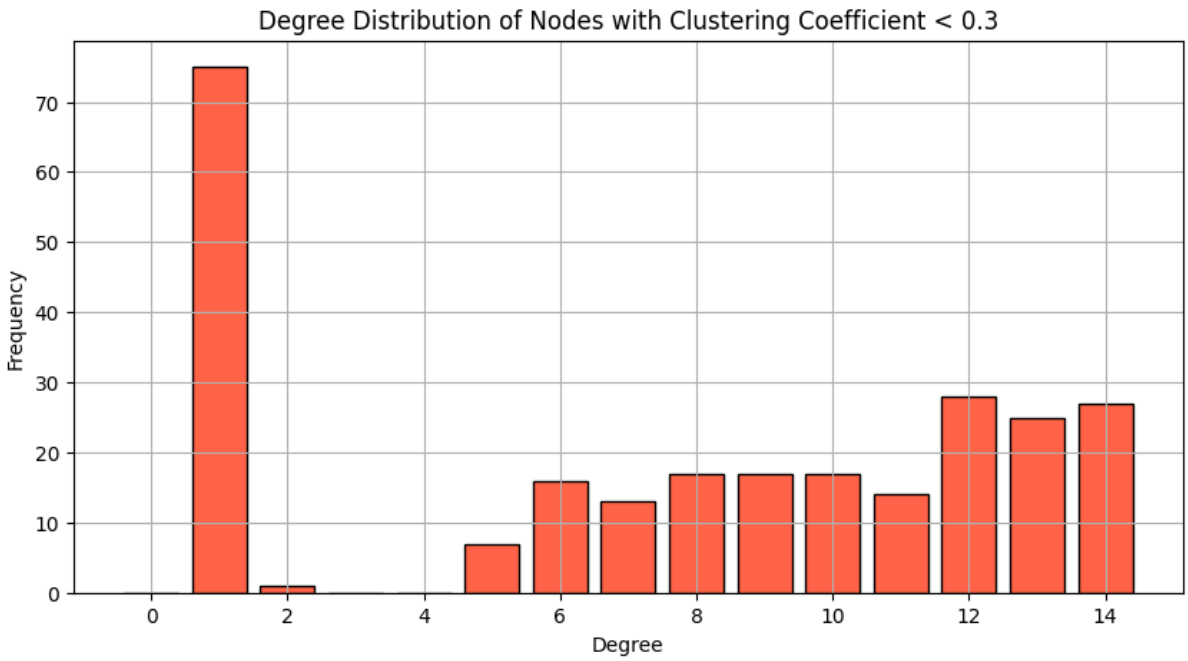

Figure 4: Degree distribution of nodes with clustering coefficient < 0.3: Isolates and structurally peripheral nodes dominate the low-clustering subset. The leftmost spike indicates many degree-1 nodes with no opportunity for triangle formation.

2.3 Community Detection Algorithm

The word "community" refers to a densely connected subgroup of the network as detected by Louvain community detection algorithm which was implemented using the `best_partition` function from the `community` Python package [2, 7]. The modularity function Q was optimised by this algorithm, which quantifies the strength of the division of a network into communities [7]. Formally, for a given partition of a graph $G = (V, E)$, the modularity is defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \Delta(c_i, c_j) \tag{3}$$

where:

- A_{ij} is the element of the adjacency matrix indicating the presence of an edge between nodes i and j,
- k_i and k_j are the degrees of nodes i and j, respectively,
- m is the total number of edges in the network,
- $\Delta(c_i, c_j)$ is the Kronecker delta, equal to 1 if nodes i and j belong to the same community, and 0 otherwise.

The Louvain algorithm proceeds in two phases:

1. Each node is initially assigned to its own community, and nodes are moved between communities to locally maximise modularity.
2. Communities are aggregated into supernodes to construct a new, smaller network on which the process is repeated. This greedy hierarchical optimization continues until no further improvement in Q is possible.

2.4 Jaccard Similarity

The Jaccard similarity between nodes u and v is defined as [9]:

$$\text{Jaccard}(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|} \tag{4}$$

where $N(u)$ and $N(v)$ denote the sets of neighbors of nodes u and v, respectively. This measure ranges from 0 (no shared neighbors) to 1 (identical neighborhoods), and it shows similarity between nodes. A high Jaccard score indicates that two nodes are more likely to share common features.

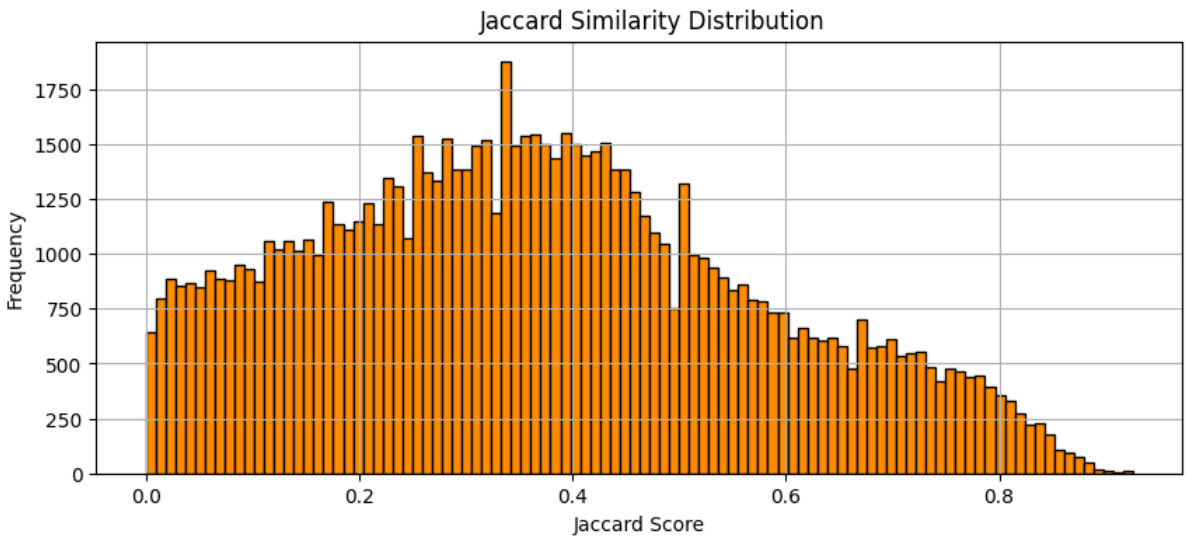

Figure 5: Jaccard similarity distribution: Histogram of Jaccard similarity scores between node pairs, used as a structural representation for social similarity. Many pairs have low overlap and their score is less than 0.3.

2.5 Intervention Strategies

2.5.1 Targeted within-community rewiring

The left behind node will connect to the most similar node within its own community. Community detection algorithm runs before each intervention round using a different random seed because the strategy should remain adaptive as the network evolved due to new edges. In here **Jaccard similarity** between nodes' neighbor sets is used to rank potential new partners for each left behind node with prioritising connections within the same community that maximise local neighborhood overlap. This increases the probability of having triangles as new connections are more likely to link individuals already adjacent to one another's social context.

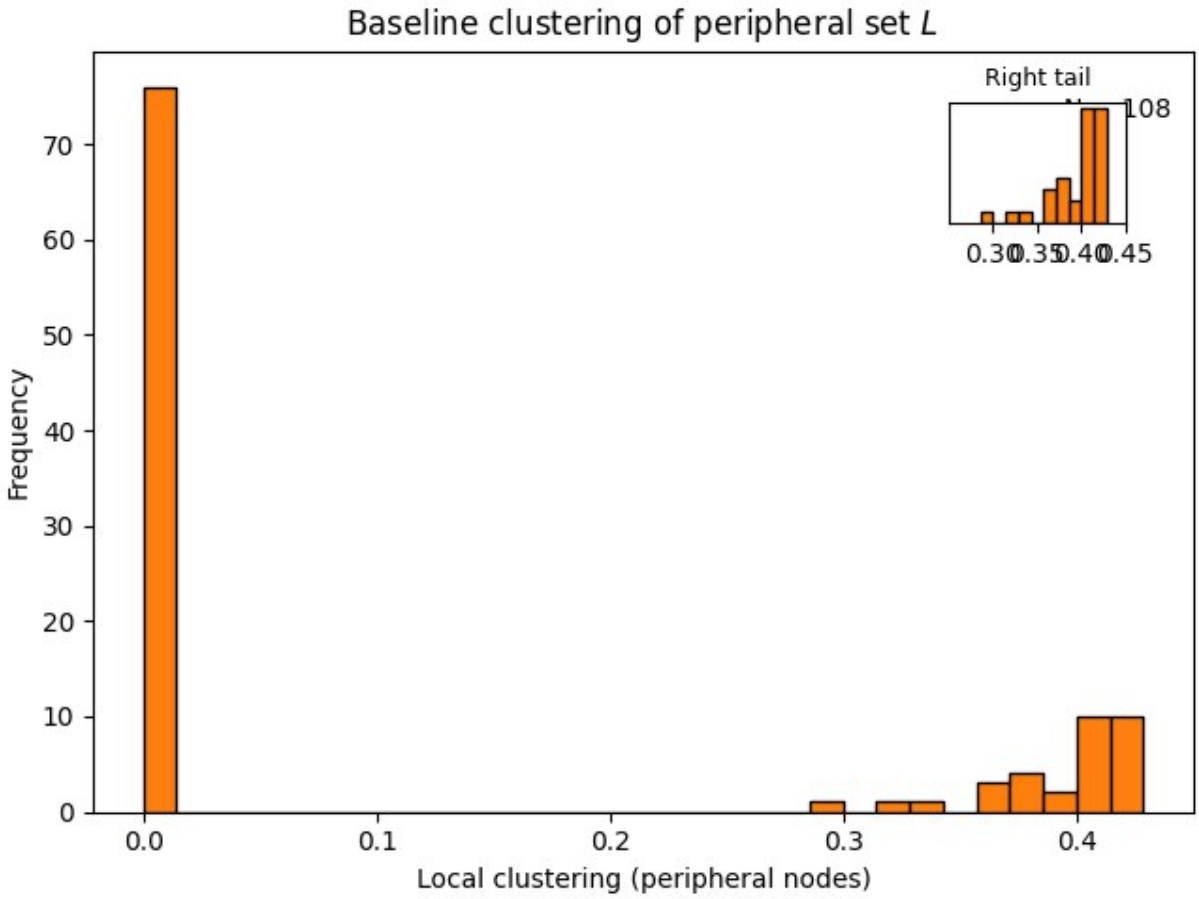

Figure 6: Baseline clustering of peripheral set L: Histogram showing the distribution of local clustering coefficients among nodes identified as peripheral (low degree and low clustering). Most nodes have clustering near zero, with a sparse right tail indicating a small number of peripheral nodes embedded in denser local structures.

2.5.2 Uniformly random rewiring

The same total number of new edges are added but each edge connects two randomly chosen nodes in the network with no bias toward left behind nodes or communities. This random condition serves as a control. These interventions are applied in parallel on identical baseline networks for a direct comparison. After each iteration the local clustering coeffcient of left behind nodes were measured to calculate the change (ΔC) because each left behind node would have a new connection in the targeted method but in the random condition some of them might have none or multiple new edges.

2.6 Connection Suggestion Algorithm

At first the `best_suggestions` function was used focusing exclusively on suggesting connections within the same community using a degree-based penalty to avoid recommending high degree nodes (hubs).

$$\text{Hub_Penalty} = \frac{\text{deg} - \text{deg_min}}{\text{deg_max} - \text{deg_min} + 10^{-9}} \tag{5}$$

$$\text{Score} = \text{jaccard}(u, v) \cdot (1 - \text{Hub_Penalty(v)}) \tag{6}$$

Later, the `suggest_many` function was introduced to improve robustness, particularly in sparse or imbalanced community structures. This version removes the degree-based penalty and implements a two-tiered candidate selection mechanism. For a given node $u \in V$ in a graph $G = (V, E)$, the algorithm first identifies intra-community candidates:

$$\mathcal{C}_{\text{in}} = \{v \in V \setminus \{u\} \mid C(v) = C(u) \wedge (u, v) \notin E\} \tag{7}$$

Each candidate $v \in \mathcal{C}_{\text{in}}$ is evaluated using the Jaccard similarity:

$$\text{Score}(u, v) = \text{Jaccard}(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|} \tag{8}$$

The top k candidates from $\mathcal{C}_{\text{in}}$ are selected based on their similarity scores. If fewer than k suggestions are found, the algorithm considers outer-community candidates:

$$\mathcal{C}_{\text{out}} = \{v \in V \setminus \{u\} \mid C(v) \neq C(u) \wedge (u, v) \notin E\} \tag{9}$$

These candidates are also scored using the same Jaccard metric and added to the result set in descending order until k suggestions are obtained.

2.7 Analysis

The primary outcome was the increase in local clustering for initially left behind nodes under each strategy. The mean ΔC (the change in local clustering) was calculated for the targeted and random method and to ensure robustness each experiment repeated across 30 independent random seeds. In each experiment both the targeted and random method were applied to the same network but with different seed. Each intervention was performed over five iteration of edge additions. In each round a fixed maximum number of edges (up to 200) were added, and new left-behind nodes were identified based on updated degree and clustering thresholds. After each iteration, the metrics including ASPL, count of left behind nodes, and local clustering coefficient were calculated again and the set of left behind nodes were updated because some nodes might have became integrated in the community with edge addition. The process continued for several rounds or until no isolated nodes remained. For statistical analysis, a paired-sample t-test was used to compare ΔC between the two strategies across the 30 trials and Cohen's d was calculated to quantify effect size.

3 Results

Across 30 simulation runs, the targeted within-community strategy produced significantly greater increases in local clustering among left behind nodes than the random edge-addition strategy.

3.1 Clustering gains

The majority of left-behind nodes had near-zero local clustering coefficients (Figure 3, median $C \approx 0$). These nodes typically had degree ≤ 2 and were left-behind nodes. After round one the mean increase in local clustering (ΔC) for left-behind nodes with targeted method was $\Delta C_{\text{target}} = 0.01818$ (SD $= 4.6 \times 10^{-5}$) and for random method was

$\Delta C_{\text{rand}} = 0.01091$ (SD $= 8.7 \times 10^{-4}$). A paired-sample t-test confirmed the difference was highly statistically significant ($t_{29} = 44.87$, $p = 2.39 \times 10^{-28}$) and effect size was extremely large (Cohen's $d = 8.19$) and exceeds conventional benchmarks ($d > 0.8$ is considered large). As shown in Figure 3, the targeted method showed higher clustering improvements across trials, while the random condition showed higher variability and lower mean gains.

3.2 Dynamics of Clustering Over Iterations

The targeted strategy continued to increase clustering among left-behind nodes (Figure 8). However, the clustering began to plateau after the third round meaning that isolated nodes gained intra-community friends. The random strategy produced early gains that declined over time and even reducing clustering in later rounds due to inter-community links that failed to reinforce local structure.

3.3 Impact on Global Efficiency

The Average Shortest-Path Length (ASPL) was a measure for global efficiency. Under the targeted method ASPL changed minimally from 3.6917 to 3.6902 after five rounds which is a 0.04% decrease (Figure 10, left panel). This is showing that the network's small-world topology was preserved by avoiding shortcuts to hubs. The random strategy decreased ASPL by 0.17% which is over four times greater than the other method [12]. This improves global connectivity but this is not exactly what we want. These results tell us that random edge addition improves global efficiency but targeted method is far superior for local intervention [12].

3.4 Coverage and Reach of Isolated Nodes

In Round 1, the targeted method added 56 edges and reduced the number of left-behind nodes from 42 to 13. By Round 3, only one left-behind node remained and that one also connected to network by round 5 (Figure 10, right panel). The random strategy failed to reach many of the same individuals. It can be seen after three rounds the median of 17 left-behind nodes remained (Figure 8, orange curve). Overall, the results show that targeted method performs better without disrupting global efficiency.

Metric	**Mean**	**SD**
delta_target	0.018176189003216627	4.572575044038232e-05
delta_random	0.010911439622599632	0.0008738756706535316
t_stat	44.86841811780365	-
p_val	2.389600477646569e-28	-
cohen_d	8.191814907564847	-

Table 1: Summary Statistics of the Metrics

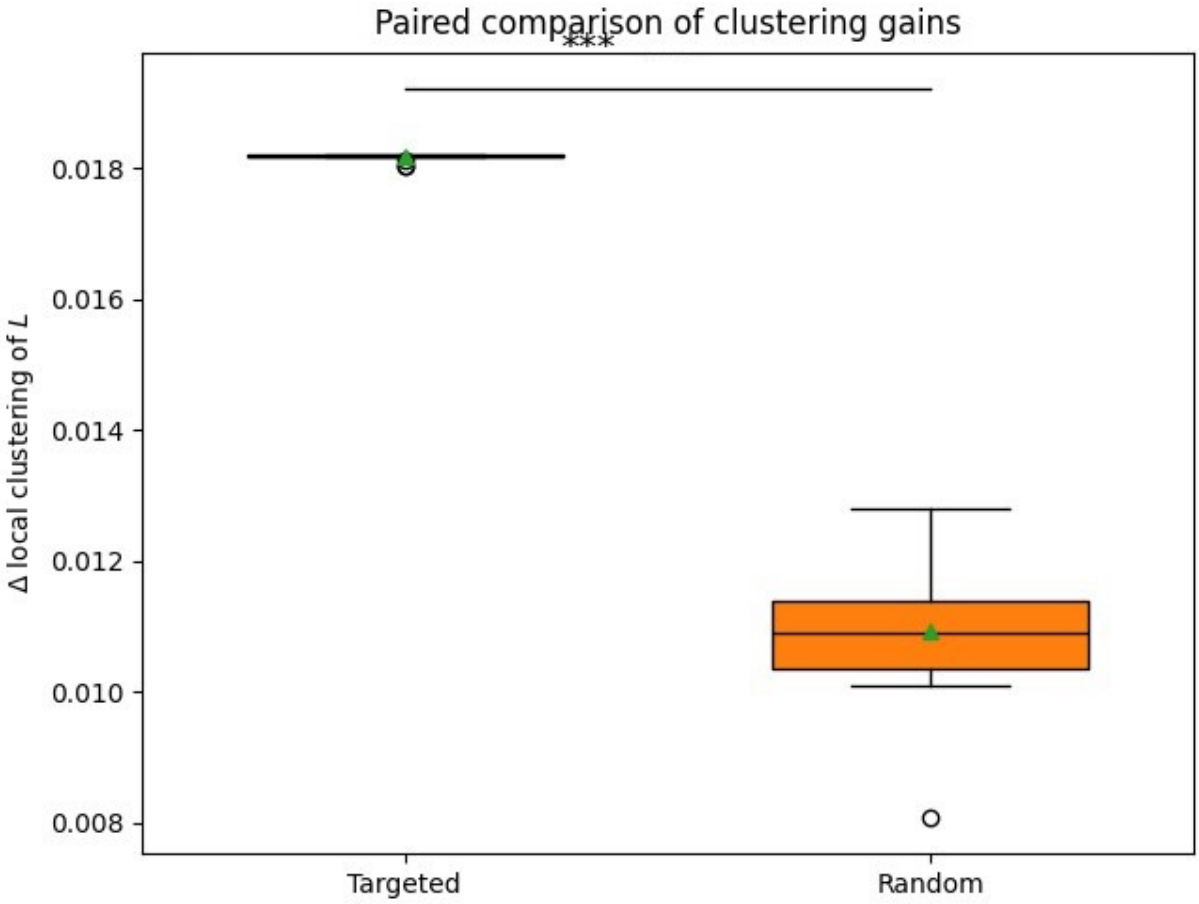

Figure 7: Paired comparison of clustering gains: Boxplot comparing the change in local clustering (ΔC) among left-behind nodes after one round of intervention. Targeted rewiring shows higher and more consistent gains than random edge addition. Statistical significance ($p = 2.39 \times 10^{-28}$) is marked, and green dots indicate the mean.

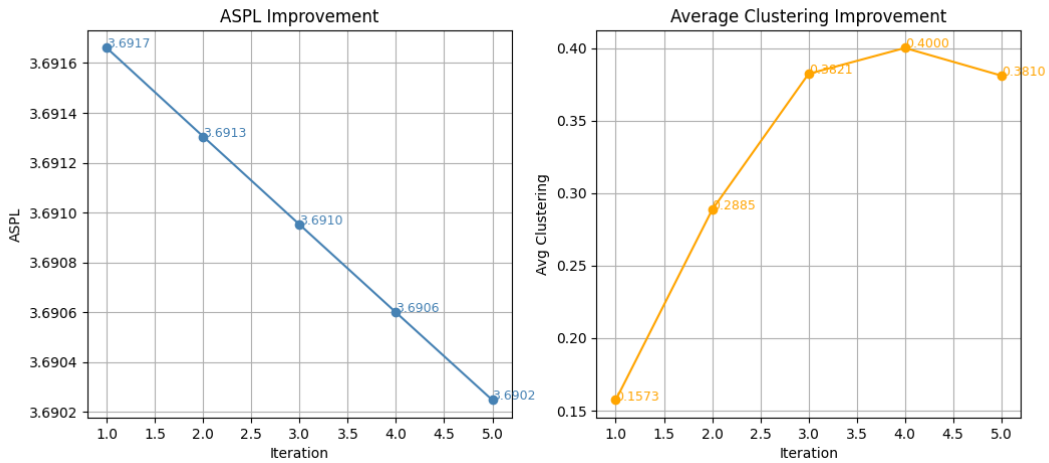

Figure 10: Global and local effects across iterations: The left panel tracks the average shortest-path length (ASPL), which decreases marginally and preserved small-world structure, while the right panel shows the mean clustering coefficient of the left-behind set rising sharply before plateauing. [12].

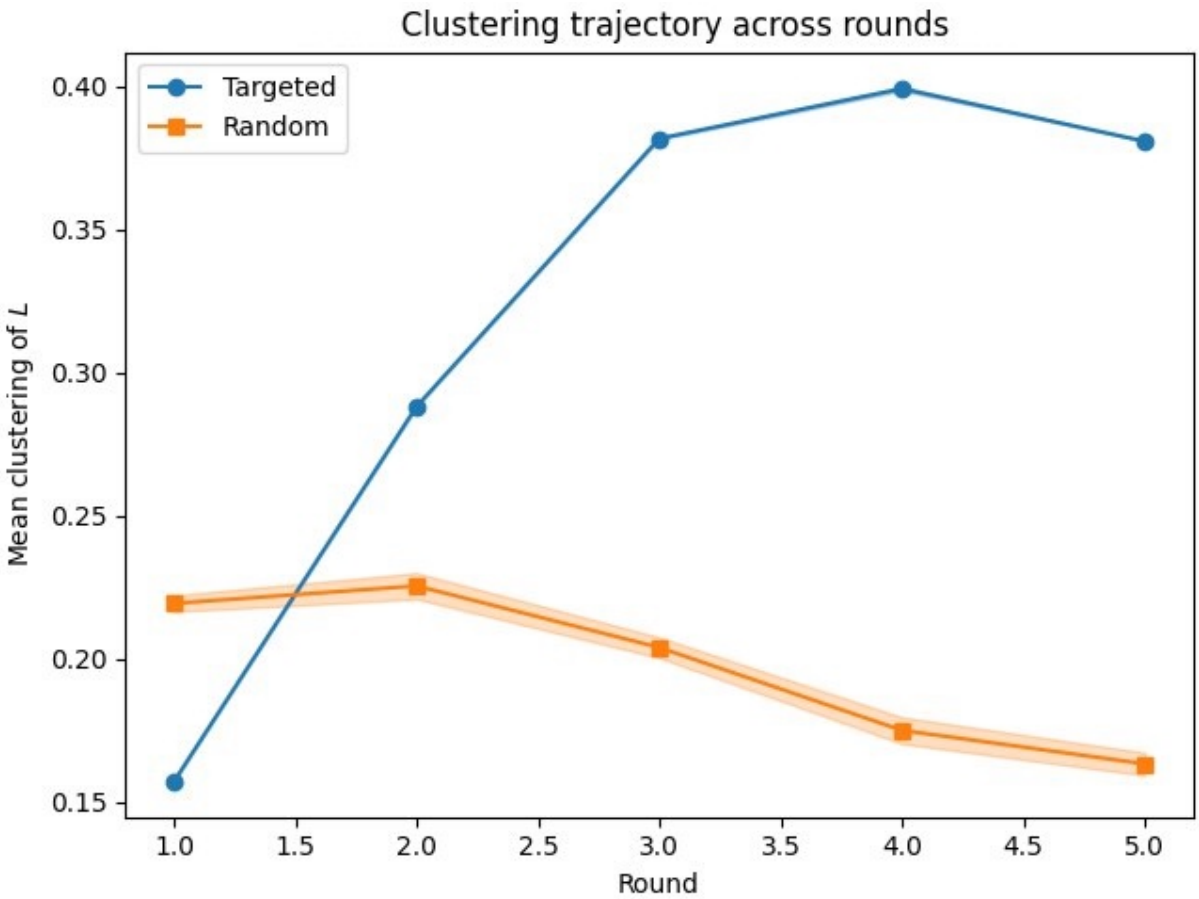

Figure 8: Clustering trajectory across rounds: Mean clustering coefficient of left behind nodes over successive iterations of edge addition. The targeted strategy (blue) shows consistent gains across rounds, while the random strategy (orange) decreases after early increases. Shaded bands represent standard error across 30 trials. These trajectories illustrate the effectiveness of targeted rewiring in embedding isolated individuals.

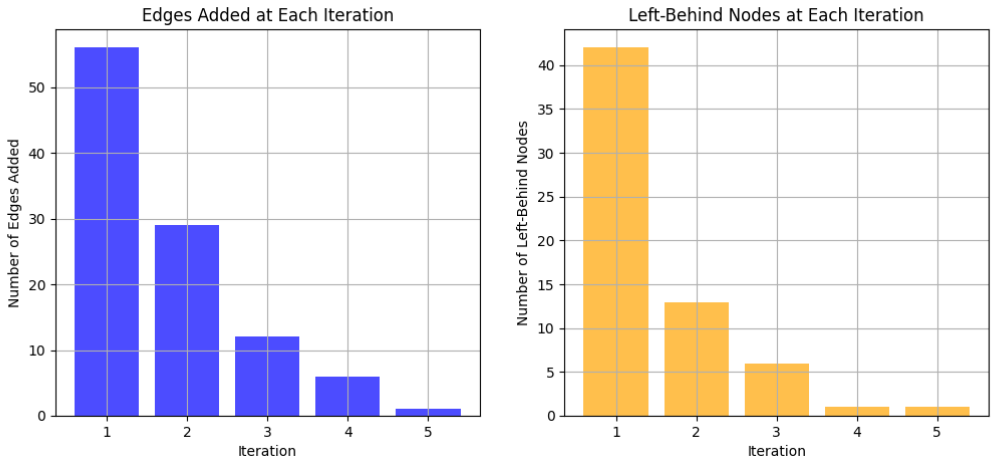

Figure 11: The left panel reports the number of new edges added during each iteration of the targeted intervention. Peripheral nodes are progressively integrated and we do not need more edges anymore in successive iterations. The right panel shows the reduction in the count of left-behind nodes, which falls from 42 to a single residual case by the fifth round.

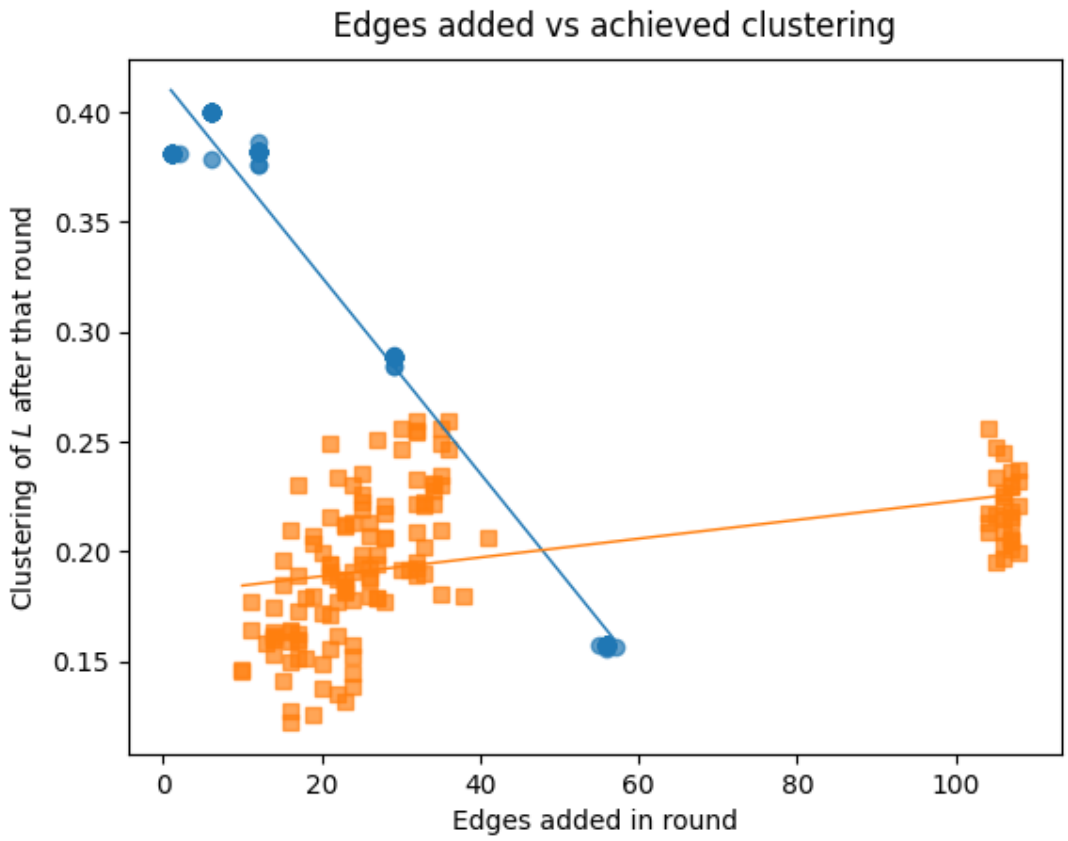

Figure 9: Edges added vs. achieved clustering: Scatterplot showing the number of new edges added in a round versus the resulting clustering of the left-behind set. In targeted strategy (blue) clustering flattens as more edges are added. Random strategy (orange) shows weaker, noisier association. Lines show linear regressions.

4 Discussion

Findings demonstrate that a targeted within-community rewiring strategy can more effectively increase the local clustering around isolated individuals than adding the same number of edges at random. If the goal is to help lonely or left behind people become embedded in friend groups where and to whom new connections are made matters [1, 12].

The new tie often closes a triangle because the isolated person's new friend may also know their old friend and boosts the local clustering coefficient. This aligns with social network theories of homophily and triadic closure which means people introduced through a mutual friend or shared group are more likely to form a cohesive triad and creating a sense of belonging in a cluster [12]. Random edges for an isolated person would connect it to a different community or to another isolated person and do not necessarily provide the features that mentioned above. These results reinforce the idea that targeting the social network periphery is a great intervention approach, as suggested by Cacioppo et al. and others [3, 1]. It is also notable that the random edge strategy performed better on a different metric called global path length showing a trade-off between local clustering vs. global efficiency in network interventions [12]. From a network science perspective, adding random long-range links is known to greatly reduce the "degrees of separation"

in a network (as in Watts-Strogatz small-world theory) and spread of information will increase [12]. If the intervention goal were to improve overall network connectivity or information flow, random linking has an advantage.

The random approach left many isolated individuals without any new friends even after multiple edge additions, simply due to chance. In a real social intervention, this would correspond to some at-risk individuals not being reached by the program and an obviously undesirable outcome. Our simulation-based approach has limitations. We used a simplified rule ("most similar node in community") to form targeted edges. In practice, identifying suitable matches for isolated people might require more nuanced criteria (shared interests, personality fit, etc.). Nonetheless, our rule captures the essence of a plausible intervention which is connecting someone to a peer in their community context and the results likely generalise to any scheme that introduces isolated individuals into existing social groups.

The absolute changes in clustering were small which may seem to suggest a limited effect but even a small structural change can correspond to a meaningful social change like one new close friendship or becoming part of a small clique [3]. These changes can compound over time and having even one or two extra friends can significantly reduce loneliness over the long term [3]

References

[1] Rebecca Band et al. "Study protocol for 'The Project About Loneliness and Social networks (PALS)': a pragmatic, randomised trial comparing a facilitated social network intervention (Genie) with a wait-list control for lonely and socially isolated people". In: *BMJ Open* 9.8 (2019). Clinical Trial Protocol, e028718. DOI: 10.1136/bmjopen-2018-028718. URL: https://bmjopen.bmj.com/content/9/8/e028718.

[2] Vincent D Blondel et al. "Fast unfolding of communities in large networks". In: *Journal of Statistical Mechanics: Theory and Experiment* 2008.10 (Oct. 2008), P10008. ISSN: 1742-5468. DOI: 10.1088/1742-5468/2008/10/p10008. URL: http://dx.doi.org/10.1088/1742-5468/2008/10/P10008.

[3] John T. Cacioppo, James H. Fowler, and Nicholas A. Christakis. "Alone in the crowd: the structure and spread of loneliness in a large social network". English. In: *Journal of Personality and Social Psychology* 97.6 (Dec. 2009), pp. 977–991. ISSN: 1939-1315. DOI: 10.1037/a0016076.

[4] Claudio Castellano, Santo Fortunato, and Vittorio Loreto. "Statistical physics of social dynamics". In: *Rev. Mod. Phys.* 81 (2 May 2009), pp. 591–646. DOI: 10.1103/RevModPhys.81.591. URL: https://link.aps.org/doi/10.1103/RevModPhys.81.591.

[5] Stephanie T Child and Leora Lawton. "Loneliness and social isolation among young and late middle-age adults: Associations with personal networks and social participation". In: *Aging Mental Health* 23.2 (Feb. 2019). Epub 2017 Nov 24, pp. 196–204. ISSN: 1364-6915 (Electronic), 1360-7863 (Print), 1360-7863 (Linking). DOI: 10.1080/13607863.2017.1399345. URL: https://doi.org/10.1080/13607863.2017.1399345.

[6] NetworkX Developers. *NetworkX Reference Documentation*. Accessed: 2025-04-10. 2022. URL: https://networkx.org/documentation/networkx-2.2/_downloads/networkx_reference.pdf.

[7] Santo Fortunato and Darko Hric. "Community detection in networks: A user guide". In: *Physics Reports* 659 (Nov. 2016), pp. 1–44. ISSN: 0370-1573. DOI: 10.1016/j.physrep.2016.09.002. URL: http://dx.doi.org/10.1016/j.physrep.2016.09.002.

[8] Mary Elizabeth Hughes et al. "A Short Scale for Measuring Loneliness in Large Surveys: Results From Two Population-Based Studies". English. In: *Research on Aging* 26.6 (2004), pp. 655–672. ISSN: 1552-7573. DOI: 10.1177/0164027504268574.

[9] Paul Jaccard. "Distribution de la Flore Alpine dans le Bassin des Dranses et dans quelques régions voisines." In: *Bulletin de la Societe Vaudoise des Sciences Naturelles* 37 (Jan. 1901), pp. 241–72. DOI: 10.5169/seals-266440.

[10] Jure Leskovec and Julian Mcauley. "Learning to Discover Social Circles in Ego Networks". In: *Advances in Neural Information Processing Systems*. Ed. by F. Pereira et al. Vol. 25. Curran Associates, Inc., 2012. URL: https://proceedings.neurips.cc/paper_files/paper/2012/file/7a614fd06c325499f1680b9896beedeb-Paper.pdf.

[11] Laura Alejandra Rico-Uribe et al. "Loneliness, Social Networks, and Health: A Cross-Sectional Study in Three Countries". In: *PLoS One* 11.1 (2016), e0145264. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0145264. URL: https://doi.org/10.1371/journal.pone.0145264.

[12] D. J. Watts and S. H. Strogatz. "Collective dynamics of 'small-world' networks". English. In: *Nature* 393.6684 (1998). Research Support, U.S. Gov't, Non-P.H.S., pp. 440–442. DOI: 10.1038/30918. URL: https://doi.org/10.1038/30918.